Book strip

Science · LAUNCH · Autumn 2 · Week 2 · SCI-LAUNCH-A2-W02-L3

Master: growth, differentiation and stem-cell evidence

I am learning to explain growth and differentiation and use evidence to repair claims about stem cells.

Success: explain the processes, compare skin and root, and repair a claim.

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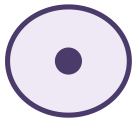
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Route card

Staff: cut these apart before the lesson. Give one route card. Other routes remain available.

Supported



Supported

Supported route card with a model cell

Match division, elongation and differentiation to more cells, longer cells and a specialised job. [3]

Use Reference A and B and the matching task sheet.

Staff I ☐ S ☐ C ☐

Independent



Independent

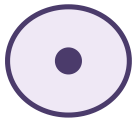
Independent route card with a model cell

Explain how division and differentiation can help replace damaged skin cells. [3]

Use Reference A and B and the matching task sheet.

Staff I ☐ S ☐ C ☐

Aspire



Aspire

Aspire route card with a model cell

A root is longer and the model shows longer cells. Explain how division and elongation could contribute. State what length alone cannot establish. [3]

Use Reference A and B and the matching task sheet.

Staff I ☐ S ☐ C ☐

Choice



Choice

Choice route card with a model cell

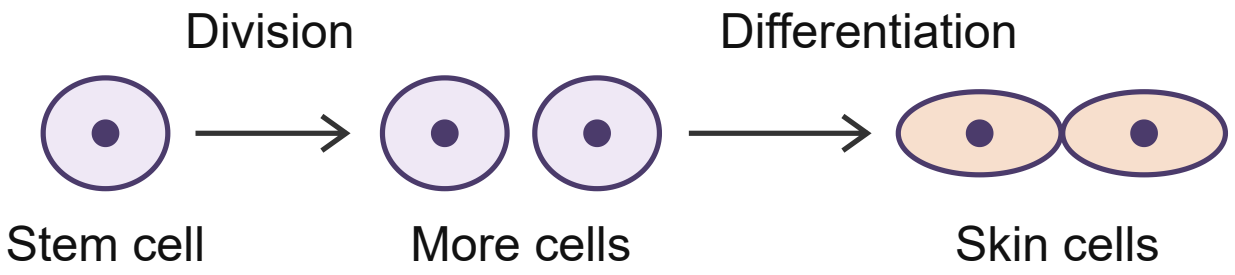
Choose division or differentiation for more cells. Choose elongation or division for longer cells. Give one reason. [3]

Use Reference A and B and the matching task sheet.

Staff I ☐ S ☐ C ☐

Made by Matt

Reference A animal cells



Division makes more cells. Some descendants specialise as skin cells.

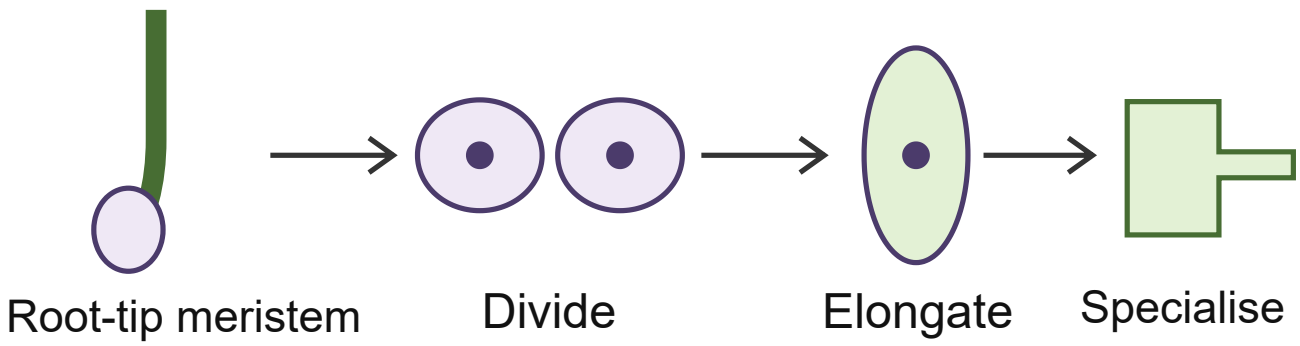
Reference only. Do not write on this sheet. Trace a feature, then explain its function.

Cell or source	Feature or range	Function or limit
Nerve cell	Long extension	Carries signals over a distance
Muscle cell	Structures that contract	Contraction produces force
Embryonic stem cell	Can give rise to many body-cell types	Not a promise of a successful treatment
Adult blood-forming stem cell	More limited range of blood-cell types	Makes replacement blood cells

The arrows show possible lineages. The drawing does not show every cell stage. Cells do not choose their own jobs. Specialised cells can share inherited DNA but use it differently.

Made by Matt

Reference B root growth



Root-tip meristem, division, elongation and differentiation in a simplified route.

Reference only. Do not write on this sheet.

Process or region	What it adds	Useful limit
Meristem	A region where cells divide	Not every root cell remains a meristem cell
Division	More cells	Number alone does not prove a specialised role
Elongation	Longer cells	Not the name for becoming specialised
Differentiation	Features suited to a job	Different cells can be at different stages

A root hair extension provides a large surface for absorption. The route is simplified. Use it to compare processes, not as a claim that every cell follows an identical sequence.

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Catch-up sheet absence or access only

Use only after absence or when an access barrier prevents joining the current task. It replaces a starting step; it adds no missed-work demand.

1. Division makes more cells. Differentiation gives cells specialised roles.

2. A stem cell can divide and give rise to specialised cell types. Adult stem-cell types have more limited ranges than embryonic stem cells.

3. In a root-tip meristem, cells divide. New cells can elongate and differentiate.

Show one link: more cells → division; a job → differentiation.

Then join the current model or You do task 1. Use Reference A or B beside the task.

verbal · written · drawn · signed · symbol / choice-selected · demonstrated / re-attempt · adult-scribed. You may pass.

Made by Matt

Choice task sheet

Select an answer and give a short reason. Use Reference A and B.

1. Choose division or differentiation for more cells. Choose elongation or division for longer cells. Give one reason. [3] Staff I ☐ S ☐ C ☐

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2. Choose both, skin only or root only for making more cells by division. Name one root feature. [2] Staff I ☐ S ☐ C ☐

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3. A root is longer. Choose growth or proof of stem-cell identity as the supported claim. Give a reason. [2] Staff I ☐ S ☐ C ☐

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